



DEPARTMENT OF TECHNICAL EDUCATION

A

PROJECT SYNOPSIS REPORT ON

" DATA ANALYTICS "

DIPLOMA IN COMPUTER SCIENCE AND ENGINEERING

PROJECT ASSOCIATES

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This is certifying that **SAFIUDDIN (180CS25701)** Students of department of computer science & engineering have successfully completed **"DATA ANALYTICS"** and submitted the synopsis in partial fulfilment of the requirement prescribed by board of technical examination, department of technical education, Bangalore for the academic year 2026-2027.

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INTRODUCTION

Data Analytics is the process of collecting, organizing, processing, and analyzing data to discover useful information and meaningful patterns. It helps organizations understand their data and make better decisions. Data Analytics uses techniques such as data cleaning, data visualization, statistical analysis, and data interpretation.

In this project, data is collected from different sources and processed to remove errors and unnecessary information. The cleaned data is then analyzed to identify trends, patterns, and relationships. The results can be presented using charts, graphs, and tables, making the information easy to understand.

The main purpose of Data Analytics is to convert raw data into useful insights that can support decision-making, improve performance, and solve real-world problems.

DATA ANALYTICS STEPS

Data Collection – Collect data from different sources such as databases, websites, surveys, or files.

Data Cleaning – Remove incorrect, duplicate, missing, or unnecessary data.

Data Organization – Arrange the collected data in a structured format such as tables or spreadsheets.

Data Processing – Process the data using suitable tools and techniques to make it ready for analysis.

Data Analysis – Examine the data to find patterns, trends, relationships, and useful information.

Data Visualization – Represent the results using charts, graphs, tables, and dashboards.

Data Interpretation – Understand the analyzed results and identify important findings.

Decision Making – Use the insights obtained from the data to support better decisions.

Reporting – Prepare a final report containing the analysis, visualizations, findings, and conclusions.

DATA CLEANING

Data Cleaning is an important step in Data Analytics that improves the quality and accuracy of data before analysis. Raw data collected from different sources often contains errors, duplicate records, missing values, incorrect formats, and unnecessary information. If these issues are not corrected, the analysis may produce inaccurate results.

In the data cleaning process, missing values are handled, duplicate records are removed, spelling mistakes are corrected, and data formats are standardized. Invalid or inconsistent data is also identified and corrected to ensure reliability. Various tools and techniques, such as spreadsheets, SQL queries, Python libraries, and data validation methods, can be used for cleaning data.

The main objective of data cleaning is to prepare high-quality, consistent, and accurate data for analysis. Clean data helps in obtaining better insights, improving decision-making, and increasing the overall effectiveness of the Data Analytics process.

DATA PROCESSING

Data Processing is an important step in Data Analytics where collected and cleaned data is transformed into a useful and understandable format. It involves organizing, sorting, filtering, and converting data according to the requirements of the analysis.

During data processing, the required data is selected and arranged systematically. Different calculations and operations are performed to identify important values and relationships. Data may also be grouped, categorized, and converted into suitable formats for further analysis.

The main purpose of data processing is to make the data ready for analysis and visualization. Properly processed data helps analysts identify patterns, trends, and useful information accurately. Tools such as Excel, SQL, Python, and Power BI can be used for effective data processing.

EXPLORATORY DATA ANALYTICS

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset before performing detailed analysis. It helps identify important patterns, trends, relationships, and unusual values present in the data.

During EDA, the data is studied using statistical methods such as mean, median, minimum, maximum, and standard deviation. Different visualizations like bar charts, pie charts, histograms, line graphs, and scatter plots are used to understand the data easily. EDA also helps detect outliers, missing values, and relationships between variables.

The main purpose of exploratory analysis is to understand the structure and quality of the data and discover useful insights. These findings help in selecting suitable analytical methods and making accurate decisions based on the data.

DATA VISUALIZATION

Data Visualization is the process of representing analyzed data in a graphical and visual form. It helps users understand large amounts of data quickly and identify important patterns, trends, comparisons, and relationships.

In this process, data is presented using bar charts, line graphs, pie charts, histograms, scatter plots, tables, and dashboards. Proper visualization makes complex information easier to understand and communicate. It also helps in identifying changes, unusual values, and important results.

The main objective of Data Visualization is to convert analytical results into clear and meaningful visuals. Tools such as Excel, Power BI, Tableau, and Python can be used to create effective visualizations. These visual representations help users understand data and support better decision-making.

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STATISTICAL ANALYSIS

Statistical Analysis is an important part of Data Analytics used to examine data and obtain meaningful information from it. It involves applying mathematical and statistical methods to understand data, identify patterns, compare values, and find relationships between different variables.

In this process, statistical measures such as mean, median, mode, minimum, maximum, range, variance, and standard deviation are calculated. Techniques such as correlation, regression, and hypothesis testing can also be used to study relationships and make conclusions from the data.

The main objective of Statistical Analysis is to understand the characteristics of the dataset and support accurate conclusions. It helps analysts identify trends, measure differences, and make data-driven decisions based on reliable results.

IDENTIFY PATTERN AND TRENDS

Identifying Patterns and Trends is an important stage of Data Analytics used to discover meaningful changes, repeated behaviors, and relationships within a dataset. By examining historical and current data, analysts can understand how different values change over time and identify useful patterns.

In this process, data is compared and analyzed using charts, graphs, statistical methods, and visualization techniques. Trends such as increasing or decreasing values, seasonal changes, frequently occurring events, and relationships between variables can be identified. These findings help in understanding the behavior of the data.

The main objective is to discover useful insights from the dataset and support better decision-making. Identifying patterns and trends also helps organizations understand performance, recognize areas that need improvement, and plan future activities based on available data.

CONCLUSION

Data Analytics is an effective process for converting raw data into meaningful and useful information. It includes important stages such as data collection, data cleaning, data processing, exploratory analysis, statistical analysis, pattern and trend identification, and data visualization.

Through these processes, large amounts of data can be examined to identify useful patterns, trends, relationships, and insights. Proper analysis helps in understanding the data clearly and supports accurate decision-making.

Overall, Data Analytics plays an important role in improving performance, solving problems, and making informed decisions. It provides a systematic approach to transforming data into valuable insights that can be used in various fields and real-world applications.

